METRIC
A-A-1818B
08 April 2020
SUPERSEDING
A-A-1818A
24 May 2005

# COMMERCIAL ITEM DESCRIPTION

# SODIUM METASILICATE, TECHNICAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) for sodium metasilicate covers material suitable for various washing, cleaning, and scouring processes, with or without soap as conditions demand, and where a moderately strong alkaline material is desired.

2. CLASSIFICATION. The sodium metasilicate shall be classified by type. The selected type of the product to be supplied shall be one of the following types as specified in the acquisition order (see 7.3(b)):

Type: I - Pentahydrate ($Na_2SiO_3 \cdot 5H_2O$)
II - Anhydrous ($Na_2SiO_3$)

3. SALIENT CHARACTERISTICS. The sodium metasilicate shall be a uniform product. The product shall conform to the following chemical and physical characteristics:

3.1 Characteristics. The sodium metasilicate, as received, shall conform to the requirements as to chemical composition and particle size prescribed in Table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DLA Aviation, VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil

AMSC N/A FSC 6810

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TABLE I. Chemical and physical requirements.

| | Type I | | Type II | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| Sodium metasilicate: | | | | |
| Calculated as $Na_2SiO_3 \cdot 5H_2O$, % by wt. | 98.0 | -- | -- | -- |
| Calculated as $Na_2SiO_3$, % by wt.[1] | -- | -- | 91.5 | -- |
| Total alkalinity, as $Na_2O$, % by wt. | 28.5 | 30.0 | 50.0 | 52.0 |
| Total silica as $SiO_2$, % by wt. | 27.5 | 29.0 | 45.0 | -- |
| Matter insoluble in water, % by wt. | -- | 0.2 | -- | 0.3 |
| Carbonates as $CO_2$, % by wt. | -- | 1.2 | -- | 2.0 |
| Particle size: | | | | |
| Material passing a 149-micron (No. 100) sieve, % by wt., maximum | -- | 10.0 | -- | 3.0 |
| Material retained on a 2.00 mm (No. 10) sieve, % by wt., maximum | -- | 10.0 | -- | 10.0 |

[1] The % by wt. of anhydrous $Na_2SiO_3$ is equal to 0.575 times the % by wt. of $Na_2SiO_3 \cdot 5H_2O$ as calculated according to ASTM D501.

3.2 Materials. The material shall be sampled and analyzed in accordance with ASTM D501, "Standard Test Methods of Sampling and Chemical Analysis of Alkaline Detergents".

3.3 Particle size. The particle size shall be sampled and analyzed in accordance with ASTM D502, "Standard Test Method for Particle Size of Soaps and Other Detergents".

3.4 Net weight. The sodium metasilicate shall be purchase by net weight but in the event of weight shortage it shall be recognized that sodium metasilicate is subject to a possible gain or loss of weight due to atmospheric or storage condition, or both. Therefore, the calculation of the net weight shall be based upon a product containing 98 percent of sodium metasilicate calculated as $Na_2SiO_3 \cdot 5H_2O$ for Type I or 91.5 percent of sodium metasilicate calculated as $Na_2SiO_3$ for Type II. The net weight of the material to be paid shall be calculated as follows:

For Type I $W = (R \times T)/98$
For Type II $W = (R \times T')/91.5$

Where:
$W$ = Net weight of the material to be paid for,
$R$ = Net weight of the material as received,
$T$ = total sodium metasilicate calculated as $Na_2SiO_3 \cdot 5H_2O$, and
$T'$ = total sodium metasilicate calculated as $Na_2SiO_3$

In the event that the calculation results in a figure greater than 100 percent, the seller waives the right to invoice the purchaser for any amount in excess of 100 percent.

4. REGULATORY REQUIREMENTS

4.1 Safety data sheet (SDS). Manufacturers shall prepare and submit a SDS in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous



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Materials Furnished to Government Activities", and meeting the requirements of 29 Code of Federal Regulations (CFR) § 1910.1200, "Occupational Safety and Health Standards: Hazard Communication".

4.2 Marking, packaging, and labeling. The product shall be labeled, packaged and marked in accordance with 49 CFR parts 100 to 185.

4.3 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance. Any certificate of analysis (COA) offered with the product to the commercial market shall be provided with each lot or batch of product. The lot or batch number and the date of inspection/analysis from the offer shall be on each COA.

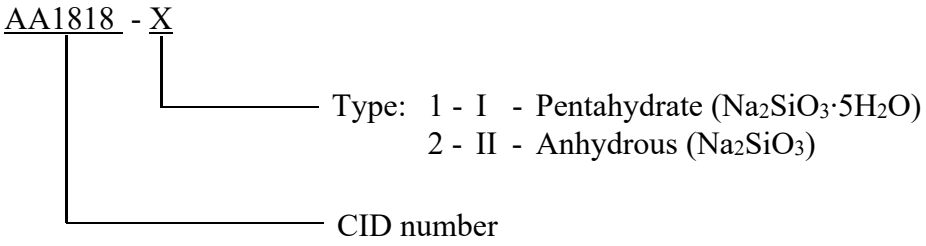
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(c)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.




Example of reference part number: AA1818 - 1 indicates Type I sodium metasilicate

7.2 Sources of documents.

7.2.1 Electronic copies of ASTM documents may be obtained from https://www.astm.org.



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7.2.2 Electronic copies of CFR documents may be obtained from https://www.ecfr.gov.

7.2.3 Electronic copies of FAR documents may be obtained from https://www.acquisition.gov/browse/index/far.

7.2.4 Electronic copies of FED-STD-313 may be obtained from https://quicksearch.dla.mil.

7.3 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Type (see 2).
c. Preservation, packing, and marking (see 6.1).

7.4 National stock numbers (NSNs). The following is a list of the NSNs assigned that correspond to this CID. This list may not be indicative of all possible NSNs associated with this document.

| NSN | Type |
|---|---|
| 6810-00-240-2121 | I |
| 6810-00-664-7062 | II |

7.5 Subject term (key word) listing.

Anhydrous
Pentahydrate
Alkaline detergents

7.6 Changes from previous issues. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

Custodians:
Army - MR
Air Force – 68
DLA - GS
NGA - MP

Review Activities:
Army - EA
Air Force - 03

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:
DLA - GS3

(Project 6810-2020-009)



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